

ETHAN SHAHROOZ MOGHADDAM

Data Scientist · Python Developer · Android & Flutter App Developer · Digital Product Creator

Available for: Full-time · Freelance / Contract · Remote · Hybrid · Open to international opportunities

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Right to Work: British citizen (passport held) · Enhanced DBS certificate (clean, available on request)

PROFILE

Imperial College London graduate in Data Science (96% average) and Python developer, with a strong portfolio spanning data analysis, machine learning, NLP, causal AI, and production-grade full-stack development. Comfortable working independently across the full project lifecycle — from raw data and model building through to deployment, documentation, and iteration.

Most recently built VoltEdge, an enterprise energy intelligence platform targeting UK and European industrial operations — featuring real-time multi-site monitoring, ML anomaly detection, causal AI root-cause analysis, and CSRD-ready ESG reporting, with a plain-English AI assistant powered by Claude. Alongside this, I independently design, build, and commercialise Android and Flutter apps and digital tools, distributing products across the Google Play Store, Gumroad, Lemon Squeezy, and Envato. Seven published apps and a catalogue of finance tools, generative art systems, and productivity products across multiple platforms. These projects sharpen real commercial instincts: pricing strategy, user feedback loops, iterative improvement, and product positioning are as much part of my toolkit as the code itself.

After a period dedicated to full-time family commitments, I returned to tech with focus and intent — building a portfolio of real-world ML and full-stack projects, shipping commercial apps and digital products to market, and continuing to grow my skills with every build. I am actively seeking a data scientist or Python developer role where I can make a meaningful contribution and continue growing within a collaborative team.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Dart (Flutter), Kotlin & Java, HTML, CSS

Libraries: pandas, NumPy, scikit-learn, spaCy, Matplotlib, Seaborn, SciPy, Plotly, FastAPI, yfinance, pdfplumber, python-docx, TensorFlow, NetworkX

Frameworks & Tools: Plotly Dash, React, Flask, Electron, Streamlit, Node.js, Firebase, Docker, Git, GitHub, Jupyter Notebook, REST APIs, Tableau, Power BI, Excel

Cloud & Deployment: AWS (ECS Fargate), GCP (Cloud Run), Railway, Render, Docker (multi-stage builds)

Machine Learning: Regression, Classification, Anomaly Detection (Isolation Forest), Causal AI (PC Algorithm), Feature Engineering, NLP, Random Forest, Logistic Regression, Gradient Boosting

Mobile Development: Android (Kotlin & Java, Google Play Store), Flutter (Dart)

AI Tools: Claude (Anthropic API — integrated into VoltEdge), ChatGPT, Gemini — integrated into development and product workflows

Product Platforms: Google Play Store, Gumroad, Lemon Squeezy, Envato

SELECTED PROJECTS

VoltEdge — Enterprise Energy Intelligence Platform

Python · FastAPI · Plotly Dash · scikit-learn · Causal AI · SQLite · Docker · Claude AI · Railway · Render

- Enterprise energy intelligence platform targeting UK and European industrial operations; monitors energy consumption across multiple sites in real time with 90 days of seeded data across 4 demo sites
- ML anomaly detection using an isolation forest model trained on each site's baseline; causal AI graph (PC algorithm via NetworkX) identifies root causes with quantified causal strength scores
- CSRD-ready ESG reporting covering Scope 2 emissions (location and market-based), GRI 302-1, and carbon intensity; plain-English AI assistant powered by Claude for natural-language querying
- JWT authentication with RBAC, SOC 2 audit trail, 903 passing tests; deployed to Render via railway.json and render.yaml configurations

GitHub: <https://github.com/ShazMoghaddam/voltedge> Live Demo: <https://voltedge-1k6q.onrender.com>

SalaryAxis — UK Salary Intelligence Platform

Python · Flask · React · pandas · NumPy · SciPy · Docker · AWS / GCP · ONS Open Data

- Production-grade UK salary intelligence platform built on ONS Annual Survey of Hours and Earnings open data, with CPI inflation adjustment, statistical analysis, and regional benchmarking
- Delivers fair salary benchmarks via a REST API and React dashboard, with regional breakdowns, gender pay gap trends, occupation distribution profiles, and CAGR analysis — GDPR-compliant, no personal data collected
- Deployable to AWS ECS Fargate or GCP Cloud Run via multi-stage Docker; full CI/CD pipeline

GitHub: <https://github.com/ShazMoghaddam/salaryaxis>

CVLens — AI CV Reader & Analyser

Python · spaCy · NLP · pdfplumber · python-docx

- NLP-driven tool to parse CVs in PDF and DOCX formats; extracts skills, detects section structure, and flags weaknesses
- Applies tokenisation, entity recognition, rule-based matching, and normalisation pipelines for consistent output
- Generates structured, human-readable feedback from unstructured text — a practical application of NLP in a real hiring context

GitHub: <https://github.com/ShazMoghaddam/CVLens>

World Map — Interactive Choropleth Country Explorer

HTML5 · CSS · Vanilla JS · SVG

- Interactive choropleth world map covering 172 countries across 7 data layers: Population Density, GDP per Capita, HDI, CO₂ per Capita, Life Expectancy, Internet Usage, and default view
- Features region and political group filtering (EU, G20, NATO, BRICS+), custom CSV/JSON data import, shareable URLs encoding selected country and data layer, iframe embedding with a full JavaScript API, and full keyboard navigation
- Single HTML file, zero dependencies, MIT licensed

GitHub: <https://github.com/ShazMoghaddam/world-map> Live Demo: <https://shazmoghaddam.github.io/world-map>

Crime Data Analysis & Classification

Python · pandas · NumPy · Matplotlib · Seaborn · scikit-learn

- End-to-end analysis of real-world crime data (2020 to present), covering cleaning, feature engineering, and exploratory analysis
- Built a Random Forest classifier to predict crime type from temporal features; evaluated with accuracy, precision, recall, and confusion matrices
- Documented assumptions, limitations, and insights clearly for both technical and non-technical audiences

GitHub: <https://github.com/ShazMoghaddam/Crime-Data-Analysis-Project>

Lending Club Loan Default Prediction

Python · pandas · NumPy · Matplotlib · Seaborn · scikit-learn

- Analysed historical loan data to identify key default risk factors across income, interest rates, and debt-to-income ratios
- Built a Logistic Regression model addressing class imbalance with stratified sampling; documented methodology and findings in a structured Jupyter Notebook

GitHub: https://github.com/ShazMoghaddam/Lending_Club_Loan_Project

Stock Tracker Dashboard & Technical Analysis

Python · Streamlit · pandas · yfinance · Plotly · Jupyter Notebook

- Built an interactive Python stock dashboard analysing AAPL, AMZN, GOOGL, and JPM using the Yahoo Finance API and Streamlit
- Implemented an SMA crossover strategy to generate buy/sell signals; visualised and documented the full workflow

GitHub: https://github.com/ShazMoghaddam/Stock_Tracker_Dashboard

ANDROID & FLUTTER APPS — GOOGLE PLAY STORE

Seven published apps across educational, utility, and creative categories, all free to download:

SearchWars — Which Gets More Google Searches?

Higher/lower guessing game built with Flutter. Players guess which of two items gets more monthly Google searches across 3,187 questions in 11 categories (Sports, Celebrity, Tech, Gaming, Geography, and more). Features 3 lives, per-category high scores, Firebase integration, and smooth card animations.

Play Store: <https://play.google.com/store/apps/details?id=com.searchwars.app>

Math Adventure – Maths for Kids

Adventure-based learning app for ages 3–11, combining storytelling and exploration with maths challenges across six themed worlds and adaptive difficulty.

Play Store: <https://play.google.com/store/apps/details?id=com.shazmoghaddam.mathadventure>

Bingo Maths

Bingo-style maths game for Android; makes practising arithmetic engaging through familiar mechanics.

Play Store: <https://play.google.com/store/apps/details?id=com.ShazMoghaddam.bingomaths>

Split Flap World Clock

Living departure board app inspired by iconic Solari split-flap displays; supports 300+ cities and multiple colour themes.

Play Store: <https://play.google.com/store/apps/details?id=com.ShazMoghaddam.splitflapworldclock>

Animated Month Wallpaper

Minimal animated live wallpaper displaying the current month with 7 language options, 5 fonts, and a live clock.

Play Store: <https://play.google.com/store/apps/details?id=com.ShazMoghaddam.animatedmonthwallpaper>

Animated Weekday Wallpaper

Minimal animated live wallpaper displaying the current day of the week with multi-language and font options.

Play Store: <https://play.google.com/store/apps/details?id=com.ShazMoghaddam.animatedweeklywallpaper>

World Builders

Cross-platform Flutter app published on the Google Play Store.

Play Store: <https://play.google.com/store/apps/details?id=com.shazmoghaddam.worldbuilders>

DIGITAL PRODUCTS

Finance tools, generative art systems, productivity utilities, and screensavers distributed across Gumroad, Lemon Squeezy, and Envato:

Market Pulse (Gumroad)

Minimal self-hosted economic calendar for traders — high-impact events, LIVE status detection, local timezone conversion, zero dependencies. Single HTML file, works fully offline.

TradeHours Pro (Gumroad)

Live global market hours tracker for 25+ markets across 6 regions. Real-time open/close countdowns, holiday calendar with half-day support, drag-to-reorder, PWA offline support, full WCAG AA accessibility.

KNOK Matrix / Mono (Gumroad)

Abstract typographic system and live digital clock where digits 0–9 are hand-composed geometric SVG artworks. Dual CSS blend mode themes (multiply / screen), template-based glyph system, single HTML file.

Typography Digits (Gumroad)

Generative word-cloud art where digits 0–9 are built from programming and data-science vocabulary. Canvas pixel masking, AABB collision detection, SVG clipPath rendering, breathing animations — zero dependencies.

Animated Live Wallpapers — Months & Weekdays (Lemon Squeezy)

Minimal animated wallpapers in HTML/CSS with 7 language options, 5 fonts, custom colour picker, and live clock. Compatible with Lively Wallpaper (Windows) and Splash (macOS), fully offline.

Aero Atlas — Terminal Time Screensaver (Envato)

Fullscreen split-flap Solari-inspired screensaver displaying live world clocks for up to 10 cities. 300+ world cities, 5 colour themes, acoustic click sounds, physics bounce animation, works offline on macOS and Windows. Built with Electron.

EXPERIENCE

Independent Digital Product Developer · Remote2022 – Present

shazmoghaddam.github.io · [Google Play Store](#) · [Gumroad](#) · [Lemon Squeezy](#) · [Envato](#)

- Built VoltEdge, an enterprise energy intelligence platform with FastAPI, Plotly Dash, causal AI anomaly detection, and CSRD-ready ESG reporting — deployed live to Render with 903 passing tests
- Independently design, develop, and commercialise seven Android and Flutter apps and a catalogue of digital tools, managing the full product lifecycle from concept and build through to publishing, pricing, and customer support
- Published apps include SearchWars (Flutter, 3,187-question higher/lower game), Math Adventure and Bingo Maths (educational maths games for children), Split Flap World Clock (300+ cities, Solari-inspired), and Animated Month & Weekday Wallpapers (multi-language, multi-platform)
- Digital product catalogue spans finance tools (Market Pulse, TradeHours Pro), generative art systems (Typography Digits, KNOK Matrix / Mono), screensavers (Aero Atlas), and animated wallpapers — across Gumroad, Lemon Squeezy, and Envato
- Apply user feedback and data-driven iteration to refine features, improve usability, and grow revenue streams across platforms

Sales Representative (Technology) · John Lewis / Currys (Agency), LondonMarch 2011 – March 2013

- Customer engagement, technical product advice, and sales support across multiple London sites
- Developed strong communication, prioritisation, and stakeholder management skills in a fast-paced retail environment

EDUCATION & CERTIFICATIONS

Imperial College London · HyperionDevJune 2025

Data Science Online Bootcamp · Average Grade: 96% · Certificate No: ES25030017561-7076

Data types, statistics, machine learning, data cleaning, and visualisation

Google Play Academy April 2026

Google Play Store Listing Certificate · Valid through April 2029 · No: 179779870

NHS Blood & Transplant 2026

50 Blood Donations Certificate

50 donations milestone — recognised by NHS Blood & Transplant

Udemy October 2025

The Data Science Course: Complete Data Science Bootcamp 2025 · 32 hours

Udemy January 2024

iOS & Swift: The Complete iOS App Development Bootcamp · 60.5 hours

Udemy October 2022 – July 2023

100 Days of Code: The Complete Python Pro Bootcamp · The Complete Python Bootcamp: Zero to Hero · 58 hours

ADDITIONAL

Notable Build: FIFA World Cup 2026 Bracket Predictor — interactive single-file web app with a full 32-team knockout bracket, 188-question trivia quiz across 6 categories, one-click PNG export, encodable pick URLs for social sharing, dark mode, and full mobile responsiveness. Zero dependencies, hosted on GitHub Pages.

GitHub: <https://github.com/ShazMoghaddam/wc2026> Live: <https://shazmoghaddam.github.io/wc2026>

Writing: Six personal essays published on Medium — on growth, identity, technology, and perspective. Writing clearly about complex or personal topics is a skill that transfers directly into documentation, stakeholder communication, and technical reporting.

Medium: <https://medium.com/@shaz.moghaddam>